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United States Patent	12404757
Kind Code	B2
Date of Patent	September 02, 2025
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Hydraulic fracturing spread and mechanisms

Abstract

A hydraulic fracturing spread system has modular high-pressure fracturing fluid manifold assemblies which couple to form a large trunk line for collecting and conveying high pressure fracturing fluid toward the well bore. Oil-hydraulic intensifier pump units integrated with the modular manifold assemblies increase the fracturing fluid pressure into the trunk line. The manifold assemblies may be configured differently from one to the other and coupleable horizontally and/or vertically.

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Family ID:	76864474
Appl. No.:	18/470215
Filed:	September 19, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240011382 A1	Jan. 11, 2024

Related U.S. Application Data

continuation parent-doc US 17762882 US 11761318 WO PCT/US2021/014004 20210119 child-doc US 18470215
us-provisional-application US 62962007 20200116

Publication Classification

Int. Cl.: E21B43/26 (20060101); F04B47/06 (20060101)

U.S. Cl.:

CPC **E21B43/2607** (20200501); **F04B47/06** (20130101);

Field of Classification Search

CPC: E21B (43/26); E21B (43/2607)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5616009	12/1996	Birdwell	N/A	N/A
12252982	12/2024	Hunter	N/A	E21B 43/2607
2009/0269250	12/2008	Panagiotou	422/129	B01F 25/23
2012/0255734	12/2011	Coli et al.	N/A	N/A
2014/0138079	12/2013	Broussard et al.	N/A	N/A
2014/0166267	12/2013	Weightman et al.	N/A	N/A
2015/0322761	12/2014	Hodgson et al.	N/A	N/A
2015/0345272	12/2014	Kajaria et al.	N/A	N/A
2018/0223640	12/2017	Keihany et al.	N/A	N/A
2019/0010793	12/2018	Hinderliter	N/A	H02P 27/047
2019/0211661	12/2018	Reckels et al.	N/A	N/A
2021/0207588	12/2020	Yeung	N/A	H02J 9/066
2022/0065237	12/2021	Cook	N/A	F04B 17/06
2022/0154775	12/2021	Liu	N/A	F16D 1/10

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106677757	12/2016	CN	N/A
2019169261	12/2018	WO	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/762,882 filed on Mar. 23, 2022, which was a U.S. National Stage Application of PCT/US21/14004 filed on Jan. 19, 2021, which claimed the benefit of U.S. Provisional Application Ser. No. 62/962,007, filed on Jan. 16, 2020, all of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to hydraulic fracturing. More particularly, the present disclosure relates to a hydraulic fracturing spread and method.

BACKGROUND

(2) Hydraulic fracturing has been used for years to extract oil and gas from shale reservoirs far

below the earth's surface. To release the oil and gas in the reservoirs, a hole is drilled straight down into the earth. A casing is then placed within the hole and cement is introduced to secure the casing in place. After the casing is secure, horizontal drilling follows and another casing is placed into the horizontal section, which allows a perforating gun to enter and puncture holes in the casing. These holes are then ready to receive fracturing fluid pumped at high pressure, causing the fractures that will release oil and/or gas. This process is a form of well stimulation referred to as hydraulic fracturing. There are many possible constituents of fracturing fluid used in various fracturing treatments, but the most common is primarily water mixed with sand and various chemicals. Pressures in excess of 15,000 psi may be required, depending on the specific well geology and the fracturing treatment.

(3) A pressure pumping arrangement for stimulating an oil well typically involves individual pumping units on truck trailers, which pump fracturing fluid for hydraulic fracturing. Flow from each individual truck is collected into a manifold and directed to a wellhead (this system is typically referred to as a “spread”). A typical fracturing treatment may require the flow of fracturing fluid from twelve trucks, each with 3000 horsepower engines pumping the fluid directly with crank-driven plunger pumps. High pressure connections are required between each pump and the manifold. These connections often present safety concerns and require expensive maintenance and replacement due to wear. With as many as twelve trucks, and often more, in a spread, there are many high-pressure connections that have to be continually checked and maintained to avoid accidents. Additionally, spreads usually require a large footprint to operate, which requires preparing a large pad around a well site.

(4) Accordingly, there is a need for a hydraulic fracturing spread that can decrease the length of high-pressure lines and decrease the number of high-pressure connections, while decreasing the typical footprint of the spread. The present disclosure seeks to solve these and other problems.

SUMMARY OF EXAMPLE EMBODIMENTS

(5) In one embodiment, a hydraulic fracturing spread comprises a hydraulic oil tank unit which supplies hydraulic oil at low pressures to manifold assemblies. The hydraulic oil, at low pressure, is sent to low pressure oil-hydraulic manifolds within one or more manifold assemblies. The hydraulic oil is then sent from the oil-hydraulic manifolds to oil-hydraulic pumper units, which increase the pressure of the hydraulic oil and pump the hydraulic oil, now at medium pressure, to intensifier units and valving, each of which are positioned in the manifold assemblies. A fracturing fluid source (e.g., a blender) sends fracturing fluid through low pressure fracturing fluid manifolds to the intensifier pump units, where the pressure of the fracturing fluid is increased to the desired pressure or PSI (e.g., high pressure). The intensifier pumps may be double-acting intensifier pumps or single-acting intensifier pumps. The high-pressure fracturing fluid is then sent to high-pressure fracturing fluid manifolds positioned on the manifold assemblies. The high-pressure fracturing fluid is delivered to a well bore to initiate the fracturing process.

(6) In one embodiment, the oil-hydraulic pumper units comprise diesel-powered pumps. In another embodiment, the oil-hydraulic pumper units comprise electric motors to drive the oil-hydraulic pumps. In another embodiment the oil hydraulic pumper units comprise turbines to drive the oil-hydraulic pumps.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a schematic diagram of a hydraulic fracturing spread;
- (2) FIG. 2 illustrates a top plan view of a hydraulic fracturing spread;
- (3) FIG. 3 illustrates a top plan view of a hydraulic fracturing spread;
- (4) FIG. 4A illustrates a top plan view of a hydraulic fracturing spread;

- (5) FIG. 4B illustrates a detailed view of manifold assemblies coupled together via a trunkline;
- (6) FIG. 5A illustrates a side elevation view of a hydraulic fracturing spread with vertically stacked manifold assemblies;
- (7) FIG. 5B illustrates a detailed view of manifold assemblies stacked vertically with high pressure fracturing fluid lines;
- (8) FIG. 6 illustrates a top plan view of a hydraulic fracturing spread;
- (9) FIG. 7 illustrates a side elevation view of a hydraulic fracturing spread;
- (10) FIG. 8 illustrates a top plan view of a hydraulic fracturing spread with horizontally coupled manifold assemblies; and
- (11) FIG. 9 illustrates a side elevation view of a hydraulic fracturing spread with vertically stacked manifold assemblies.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

- (12) The following descriptions depict only example embodiments and are not to be considered limiting in scope. Any reference herein to “the invention” is not intended to restrict or limit the invention to exact features or steps of any one or more of the exemplary embodiments disclosed in the present specification. References to “one embodiment,” “an embodiment,” “various embodiments,” and the like, may indicate that the embodiment(s) so described may include a particular feature, structure, or characteristic, but not every embodiment necessarily includes the particular feature, structure, or characteristic. Further, repeated use of the phrase “in one embodiment,” or “in an embodiment,” do not necessarily refer to the same embodiment, although they may.
- (13) Reference to the drawings is done throughout the disclosure using various numbers. The numbers used are for the convenience of the drafter only and the absence of numbers in an apparent sequence should not be considered limiting and does not imply that additional parts of that particular embodiment exist. Numbering patterns from one embodiment to the other need not imply that each embodiment has similar parts, although it may.
- (14) Accordingly, the particular arrangements disclosed are meant to be illustrative only and not limiting as to the scope of the invention, which is to be given the full breadth of the appended claims and any and all equivalents thereof. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation. Unless otherwise expressly defined herein, such terms are intended to be given their broad, ordinary, and customary meaning not inconsistent with that applicable in the relevant industry and without restriction to any specific embodiment hereinafter described. As used herein, the article “a” is intended to include one or more items. When used herein to join a list of items, the term “or” denotes at least one of the items, but does not exclude a plurality of items of the list. For exemplary methods or processes, the sequence and/or arrangement of steps described herein are illustrative and not restrictive.
- (15) It should be understood that the steps of any such processes or methods are not limited to being carried out in any particular sequence, arrangement, or with any particular graphics or interface. Indeed, the steps of the disclosed processes or methods generally may be carried out in various sequences and arrangements while still falling within the scope of the present invention.
- (16) The term “coupled” may mean that two or more elements are in direct physical contact. However, “coupled” may also mean that two or more elements are not in direct contact with each other, but yet still cooperate or interact with each other.
- (17) The terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments, are synonymous, and are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including, but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes, but is not limited to,” etc.).
- (18) As previously discussed, there is a need for a hydraulic fracturing spread that can decrease the number of pressure pumps, decrease the number of high-pressure connections, and that decreases

the typical hydraulic fracturing spread footprint. As will be appreciated, the hydraulic fracturing spread disclosed herein solves these problems and others.

(19) A pressure pumping arrangement for stimulating an oil well typically has individual pumping units on truck trailers, which pump fracturing fluid for hydraulic fracturing. Pressures in excess of 15,000 psi may be required, depending on the specific well geology and the fracturing treatment. Flow from each pumping unit is collected in a manifold and directed to the wellhead. The fracturing treatment may require flow of oil from numerous trucks (e.g., **12** trucks), each having their own pumping unit. High pressure connections are required between each pump and the manifold. These high-pressure lines present safety concerns and require expensive maintenance. This typical spread also requires a large footprint due to the many trucks with pumps and the trucks for transport.

(20) Generally, a hydraulic fracturing spread system described herein comprises manifold assemblies with pressure intensifier pumps. Oil-hydraulic pumps may be coupled separately from a pressure intensifier, where hydraulic oil at medium pressure (e.g., 6000 PSI) drives a piston which pumps the fracturing fluid via a smaller diameter plunger piston. In the pressure intensifier units, the difference of the pressure-affected area of the oil hydraulic piston and the fluid pumping plunger represents an intensification ratio (e.g., 3:1) which, in turn, provides an increase in pressure of the fracturing fluid pumped by the plunger (e.g., 18,000 PSI). It will be appreciated that, due to the aid of the pressure intensifier units, integrated into manifold assemblies which collect and transport the high-pressure fracturing fluid toward the well bore, less length of high-pressure lines is required and fewer high-pressure connections are made on the well site, thereby decreasing safety concerns, increasing efficiency due to less downtime for maintenance. Furthermore, consolidating more horsepower on each oil-hydraulic pumping unit may reduce the number of transportation units and decrease the footprint size required for the spread. Additionally, the manifold assemblies allow a user to easily transport a whole hydraulic fracturing spread from site to site.

(21) Referring to the diagram illustrated in FIG. **1**, in one embodiment, a hydraulic fracturing spread system **100** comprises a hydraulic oil tank unit **102** (hydraulic oil source) which supplies hydraulic oil at first, low pressures to manifold assemblies **104A-104C**, which are coupled together. More specifically, hydraulic oil at low pressure is sent, for example, via low pressure oil lines (shown as lines A) to low pressure oil-hydraulic manifolds **106A-106C** within each of the manifold assemblies **104A-104C**. The hydraulic oil is then sent from the oil-hydraulic manifolds **106A-106C** at low pressure (lines A) to oil-hydraulic pumper units **108A-108C**, which increase the pressure of the hydraulic oil. It will be appreciated that the low pressure oil-hydraulic manifolds **106A-106C** are interposed between the hydraulic oil tank unit **102** and the one or more oil-hydraulic pumper units **108A-108C**. The hydraulic oil is then pumped at a second, medium pressure flow (shown as lines B) to intensifier units and valving **110A-110C**, each of which are positioned in the manifold assemblies **104A-104C**. This is a distinction from the prior art, which requires high-pressure connections from the oil-hydraulic intensifiers and the pumper units. By removing these high-pressure connections, safety is increased, downtime and maintenance is reduced, among other things, which is a considerable improvement over the prior art.

(22) A fracturing fluid source **112** (e.g., a blender) sends fracturing fluid through first, low pressure (lines C) to first fracturing fluid manifolds **114A-114C** (low pressure manifolds) within the manifold assemblies **104A-104C**. These low-pressure manifolds **114A-114C** send the fracturing fluid at low pressure (lines C) to the intensifier pump units and valving **110A-110C**, where the pressure for the fracturing fluid is increased to the desired PSI. It will be appreciated that the low-pressure manifolds **114A-114C** may be interposed between the fracturing fluid source **112** and the intensifier pump units and valving **110A-110C**. It further will be appreciated that the intensifier pump units and valving **110A-110C** use clean oil-hydraulic pressure to drive and increase the pressure of the fracturing fluid to a second, high pressure fracturing fluid. In some embodiments, as

later described, the intensifier pump units and valving **110A-110C** may be double-acting intensifier pumps or single-acting intensifier pumps.

(23) The second, high-pressure fracturing fluid is then sent to second fracturing fluid manifolds **116A-116C** (high-pressure manifolds), positioned on the manifold assemblies **104A-104C**, via high pressure flow lines (lines D) from the intensifier pump units **110A-110C**. The high-pressure flow (lines D) of fracturing fluid is then sent to a well bore **118** to initiate the fracturing process. In some embodiments, the high-pressure flow (lines D) is transported through a large diameter pressure trunk line (shown in FIGS. 4A-5B). Accordingly, the high-pressure fracturing fluid is sent from one manifold assembly to the next via the high-pressure trunk line **131** (FIGS. 4A-5B). In some embodiments, the trunk lines couple the manifold assemblies **104A-104C** to each other. While three manifold assemblies **104A-104C** are shown, it will be appreciated that one or more assemblies may be used in the hydraulic fracturing spread system **100**. The term “well bore” as used to describe **118** is a generalization of the connection to the formation for fracturing. It will be appreciated there are multiple components after the manifold (or “missile”) in a typical fracturing spread. For example, instrumentation, valves, a “zipper” manifold which connects to multiple well heads, the well heads themselves, tubing and casing may be part of the connection to the wellbore.

(24) Additionally, each of the manifold assemblies **104A-104C** as illustrated in FIG. 1 comprise low pressure oil-hydraulic manifolds **106A-106B**, intensifier pump units and valving **110A-110C**, low-pressure fracturing fluid manifolds **114A-114C**, and high-pressure fracturing fluid manifolds **116A-116C**. However, it will be understood that numerous configurations of the manifold assemblies may be envisioned. For example, one manifold assembly may comprise a generator, a hydraulic oil tank, and a portion of trunk line while another manifold assembly may comprise a dual-action intensifier pump, oil-hydraulic pumper units, and a portion of the trunk line, with the rest of the manifold assemblies being configured as shown in FIG. 1. The various manifold assembly configurations may then be coupled together, via the trunk line (e.g., **131** in FIGS. 4A-5B), to create a hydraulic fracturing spread system.

(25) As shown in FIGS. 2-5B, the oil-hydraulic pumper units **108** may be mounted separately from the manifold assemblies **104** (which comprise the intensifier units and valving **110** (shown in FIG. 1)). In the oil-hydraulic pumper units **108**, hydraulic oil at medium pressure (e.g., 6000 PSI) drives a piston **119** which pumps fracturing fluid via a plunger piston (e.g., smaller diameter). The pumper units **108** comprise a pump actuator **120** (e.g., engine, motor, turbine, etc.) that drives a pump **109** of the oil-hydraulic pumper units **108**. The pump actuator **120** may be coupled with the oil-hydraulic pumps **109** on a truck trailer **122**, skid, or any other location. In one embodiment, multiple pump actuators **120** may be coupled to a truck trailer **122**. Thus, the same amount of hydraulic horsepower can be condensed to a smaller footprint (e.g., 4 truck trailer units instead of 12). The example oil-hydraulic pumper units **108** shown in FIGS. 2-3 are illustrated as large piston engines (the pump actuator **120**) with a plurality of hydraulic pumps **109** attached to each engine via a geared transfer case. However, it will be appreciated that one or more pump actuators each driving one or multiple pumps can be unitized for a specific application. Oil-hydraulic pumper units **108** refer to a system which increases the pressure of the hydraulic oil and pumps the hydraulic oil at medium pressure.

(26) While the truck-mounted example is shown and described above, it will be appreciated that, in some embodiments, the oil-hydraulic pumping units **108** comprising one or more pump actuators **120** may be mounted to skids and stacked or coupled together with other oil-hydraulic pumping units, and/or with the intensifier pumps **110**. Each pump actuator **120** may have multiple oil-hydraulic pumps **109** driven by a gear train (as shown), or one large oil-hydraulic pump may be driven by the pump actuators **120**. In some embodiments, the oil-hydraulic pumps **109** are each individually driven by a separate pump actuator **120**, such as a motor, engine, turbine, etc. In some embodiments, the pump actuators **120** are internal combustion engines (e.g., diesel or natural gas) or turbine engines. In some embodiments, the pump actuators **120** may be electric motors. The

electric motors may be used to drive the oil-hydraulic pumps **109** on the pumper units **108**. In some embodiments, a large, central generator (e.g., diesel or turbine) may produce power to drive a plurality of the electric motors (i.e., pump actuators **120**) of the oil-hydraulic pumper units **108**. Alternatively, turbine or piston engines can be used to power the oil-hydraulic pumps **109**, or any other suitable mechanism or prime mover.

(27) Intensifier units and valving **110** may be driven by the oil-hydraulic pumper units **108**, or flow from multiple oil-hydraulic pumps may be consolidated to drive individual intensifiers. In particular, the intensifier units may be driven by hydraulic oil sent through oil-hydraulic lines and connections **124** (e.g., medium pressure lines). It will be appreciated that using medium pressure lines **124**, and fewer of them than is commonly used on spreads, that connect the oil hydraulic pumps **109** to the pressure intensifier pumps **110** decreases maintenance and decreases the likelihood of accident. In contrast, systems in the prior art use many high-pressure fracturing fluid connections and lines that have to be maintained to avoid accidents. In some embodiments, a double intensifier pump may be used, such as the pump disclosed in U.S. Pat. No. 5,879,137, issued on Mar. 9, 1999, which is incorporated herein by reference. The intensifier pump shown in the prior art is double-acting, where fluid is pumped in both directions of the hydraulic piston's travel. The hydraulic spread system **100** may comprise a single- or double-acting intensifier pump as shown in the prior art. In some embodiments, the oil-hydraulic pumper units **108** and the intensifier units and valving **110** may be mounted together, forming a single unit in the manifold assemblies **104**. However, it will also be appreciated that the intensifier units and the valving (e.g., control valves) may be separated from one another. For example, the valving may be located on the oil-hydraulic pumper unit or in any other location that allows for functionality of the valves.

(28) In one embodiment, electronically controlled pumps could be implemented to control flow. Additionally, control valves for oil-hydraulic pumper units **108** may be on the pumping units **108** and/or the intensifier units **110**. The hydraulic oil tank unit **102** and cooling may be on each of the truck beds **122** or separate from the oil-hydraulic pumper units **108**. For example, one large tank and cooling system can be on a separate truck trailer or skid and shared between multiple hydraulic pump units **108**. Accordingly, a large tank can service a plurality of oil-hydraulic pumper units **108**. Further, in some embodiments, the manifold assemblies **104** may distribute hydraulic oil flows between themselves (as shown in FIG. 1). For example, low pressure fluid could flow (along lines A) from a large central tank **102** near the blender unit, through a series of connections (lines A) between manifold assemblies **104** to supply oil-hydraulic pumper units **108**. Exhausted low pressure hydraulic oil leaving intensifiers **110** and control valves can then return through a series of connections (lines A) between manifold assemblies **104** back to the central tank **102**.

(29) As previously discussed, fracturing fluid from the fracturing fluid source **112** (FIG. 1) feeds the pressure intensifier manifolds **114** and the intensifier pumps **110** at relatively low pressure via a low-pressure intake line **126** (FIG. 2). High-pressure fracturing fluid is then pumped to a wellhead **118** via a high-pressure output line **128** at a higher pressure due to the intensifier pumps **110**. Referring to FIGS. 4A, 4B, 5A, and 5B, the high-pressure fracturing fluid can be moved through high pressure fracturing fluid lines **129** to a trunk line **131** coupled to each of the manifold assemblies **104**. Additionally, FIGS. 4A and 4B show manifold assemblies **104** coupled horizontally to each other via the trunk lines **131**. In particular, the trunk lines **131** may comprise a joint **133**, which allows each trunk line section on each manifold assembly **104** to be coupled to each other. The joint **133** may be a flange with a seal that receives bolts, or any other type of coupler. Furthermore, FIGS. 5A and 5B show manifold assemblies **104** coupled and stacked vertically with the high-pressure fracturing fluid lines **129** connecting the bottom manifold assemblies to the top manifold assemblies through the intensifier pumps **110** to consolidate the flow from the bottom and top manifolds into the trunk line **131**. Accordingly, the trunk line **131** runs through each manifold assembly **104**. Thus, the oil-hydraulic pumper units **108** may be adjacent to each manifold assembly **104**. When multiple manifold assemblies **104** are coupled

together, the length of the trunk line **131** increases as well as the number of intensifier units and valving **110**. It will be appreciated that having manifold assemblies **104** comprising trunk lines **131** to move the fracturing fluid and intensifier units and valving **110** to increase the pressure of the fluid creates a smaller, easier to maintain layout. Additionally, the manifold assemblies **104** are easily transported and connected with fewer connections than hydraulic fracturing layouts in the prior art, which increases efficiency, decreases energy loss, and decreases the number of high-pressure lines. In some embodiments, the manifold assemblies **104** may comprise a frame/skid **130**, allowing the manifold assemblies **104** to be placed directly on the ground. However, it will also be appreciated that components of hydraulic fracturing spreads are commonly transported and used as skids, integrated with a vehicle or as a vehicle trailer assembly. All parts of this system can have any of those options as well for the specific application, or any other means of transporting and placing into service.

(30) To supply the pressure and flow requirements for a particular pumping treatment, multiple manifold assemblies **104** with multiple intensifier pump units **110** may be used. As shown in FIGS. 7-8, the manifold assemblies **104** may be coupled horizontally. As shown in FIGS. 6 and 9, in one embodiment, the manifold assemblies **104** may be stacked vertically. Additionally, in some embodiments, the hydraulic spread system **100** may comprise a combination of both vertically stacked manifold assemblies **104** and horizontally coupled manifold assemblies **104**. It will be appreciated that any other configuration of the manifold assemblies **104** is within the parameters of the hydraulic spread system **100**. The manifold assemblies **104**, whether in a vertical or horizontal coupling position, may couple to each other with alignment hardware, such as tapered alignment pins on the skid **130**, which may allow for ease of attaching the fluid connections between manifold assemblies **104**. For example, API flange connections between manifold assemblies **104** may allow few large diameter connections rather than multiple small diameter connections with swivel joints.

(31) The hydraulic fracturing spread system **100** may condense the footprint further than is possible for truck-mounted units. For example, the system **100** allows multiple manifold assemblies to be stacked. With off-shore applications, for example, footprint must be minimized and ease of loading and unloading equipment by crane is needed. Accordingly, the system **100** decreases footprint size for both land and off-shore applications. Not only is the footprint reduced but the number of connections and lengths required to get the hydraulic fracturing fluid to the high-pressure pumps and to the well bore is reduced. While oil-based fluids were used as examples throughout, the present disclosure is not so limited, and other pressurizable fluids (e.g., water, water and glycol, etc.) may be used without departing herefrom.

(32) It will also be appreciated that systems and methods according to certain embodiments of the present disclosure may include, incorporate, or otherwise comprise properties or features (e.g., components, members, elements, parts, and/or portions) described in other embodiments. Accordingly, the various features of certain embodiments can be compatible with, combined with, included in, and/or incorporated into other embodiments of the present disclosure. Thus, disclosure of certain features relative to a specific embodiment of the present disclosure should not be construed as limiting application or inclusion of said features to the specific embodiment unless so stated. Rather, it will be appreciated that other embodiments can also include said features, members, elements, parts, and/or portions without necessarily departing from the scope of the present disclosure.

(33) Moreover, unless a feature is described as requiring another feature in combination therewith, any feature herein may be combined with any other feature of a same or different embodiment disclosed herein. Furthermore, various well-known aspects of illustrative systems, methods, apparatus, and the like are not described herein in particular detail in order to avoid obscuring aspects of the example embodiments. Such aspects are, however, also contemplated herein.

(34) Exemplary embodiments are described above. No element, act, or instruction used in this description should be construed as important, necessary, critical, or essential unless explicitly

described as such. Although only a few of the exemplary embodiments have been described in detail herein, those skilled in the art will readily appreciate that many modifications are possible in these exemplary embodiments without materially departing from the novel teachings and advantages herein. Accordingly, all such modifications are intended to be included within the scope of this invention.

Claims

1. A hydraulic fracturing spread layout system, comprising: a plurality of oil-hydraulic pumping units, each oil-hydraulic pumping unit comprising one or more pump actuators and one or more oil-hydraulic pumps; one or more first manifold assemblies, the one or more first manifold assemblies comprising at least one intensifier pump and a trunk line; one or more second manifold assemblies, the one or more second manifold assemblies comprising at least one intensifier pump and at least one high-pressure fluid line configured to couple to the trunk line of the one or more first manifold assemblies to consolidate high pressure fracturing fluid into the trunk line.
 2. The hydraulic fracturing spread layout system of claim 1, wherein the plurality of first and second manifold assemblies are vertically coupled to one another.
 3. The hydraulic fracturing spread layout system of claim 1, wherein the plurality of first and second manifold assemblies are adjacent and horizontally coupled to one another.
 4. The hydraulic fracturing spread layout system of claim 1, wherein at least one of the plurality of first and second manifold assemblies are horizontally coupled and at least one of the plurality of first and second manifold assemblies are vertically coupled.
 5. The hydraulic fracturing spread layout system of claim 1, wherein the one or more pump actuators comprise a diesel engine, a natural gas engine, turbine, or motor.
 6. The hydraulic fracturing spread layout system of claim 1, wherein the plurality of manifold assemblies each comprise one or more skids.
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